For my Heart Disease Prediction & Signal Analysis midterm project, I used Google Gemini in my Kaggle workflow to help troubleshoot syntax errors, structure advanced ensemble code, and figure out why my models were stuck around a 0.48 to 0.51 ROC-AUC score even after heavy tuning. What I really learned from this project is that no amount of algorithm complexity—whether it is applying SMOTE, adjusting class weights in XGBoost, running GridSearchCV, or building a soft-voting ensemble with XGBoost, LightGBM, and Random Forest—can pull a prediction out of data if the features have no real statistical connection to the target. By digging into the Pearson correlations and running SHAP TreeExplainer plots, I proved that all 20 patient features had near-zero linear correlation (within ±0.02) with the raw Heart Disease Status column, and that Stress Level had almost no impact (0.016) compared to clinical biomarkers like BMI, Homocysteine Level, and CRP Level. I also learned how important it is to split raw encoded data before fitting StandardScaler so test data never leaks into the training phase. Instead of just settling for a 0.50 random-guess model when my baseline Random Forest failed to catch the minority class, I took control of the notebook and built a 16-step investigation to test every angle. To make this project my own, I engineered custom cardiovascular risk features—including a Metabolic Risk Score, a Cardio Strain Index, an Inflammatory Vascular Score, and a consolidated Lipid Risk Score—while fixing duplicate column issues from degree-2 polynomial interactions and cleaning up feature names for LightGBM. Finally, I designed a custom signal verification experiment from scratch using realistic clinical weights for Age, BMI, Blood Pressure, Cholesterol, Homocysteine, Smoking, and Diabetes, paired with stratified shuffling, which proved my pipeline worked as intended when it jumped from ~0.50 on the noisy synthetic data to 0.9948 with my Soft-Voting Ensemble and 1.0000 ROC-AUC (100% accuracy, precision, and recall) with my final high-capacity Logistic Regression model.